faster it dissipates. Any primordial black hole in the final stages of decay today would have to weigh about as much as a mountain. All the smaller ones are gone. Since the abundance—to say nothing of the existence—of primordial black holes depends on what happened in the earliest moments after the Big Bang, no one can be sure that there are any to be found; we certainly can't be sure that any lie nearby. Not very restrictive upper limits on their abundance have been set by the failure so far to find short gamma ray pulses, a component of the Hawking radiation.

In a separate study, G. E. Brown of Caltech and the pioneering nuclear physicist Hans Bethe of Cornell suggest that about a billion nonprimordial black holes are strewn through the Galaxy, generated in the evolution of stars. If so, the nearest may be only 10 or 20 light-years away.

If there are black holes within reach—whether they're as massive as mountains or as stars—we will have amazing physics to study firsthand, as well as a formidable new source of energy. By no means do I claim that brown dwarfs or primordial black holes are likely within a few light-years, or anywhere. But as we enter interstellar space, it is inevitable that we will stumble upon whole new categories of wonders and delights, some with transforming practical applications.

I do not know where my train of argument ends. As more time passes, attractive new denizens of the cosmic zoo will draw us farther outward, and increasingly improbable and deadly catastrophes must come to pass. The probabilities are cumulative. But, as time goes on, technological species will also accrue greater and greater powers, far surpassing any we can imagine today. Perhaps, if we are very skillful (lucky, I think, won't be enough), we will ultimately spread far from home, sailing through the starry archipelagos of the vast Milky Way Galaxy. If we come upon anyone else—or, more likely, if they come upon us—we will harmoniously interact. Since other space-faring civilizations are likely to be much more advanced than we, quarrelsome humans in interstellar space are unlikely to last long.

Eventually, our future may be as Voltaire, of all people, imagined: